MIL-P-25770A(ASG)
27 NOVEMBER 1961

Superseding
MIL-P-25770(USAF)
8 January 1959

MILITARY SPECIFICATION

PLASTIC MATERIALS, ASBESTOS BASE, PHENOLIC RESIN, LOW OR HIGH PRESSURE LAMINATES

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers asbestos base, phenolic resin, low or high pressure laminated plastic materials for aircraft structural parts, including all external plastic parts (see 6.1).

1.2 Classification.- The asbestos base, phenolic resin, low or high pressure laminated plastic materials shall be classified by the grade as listed in tables I, II, and III. The laminated materials shall be of the following types and classes, as specified (see 6.2). The requirements of tables I, II, and III shall be applicable to each type.

Type I - General purpose
Type II - Heat resistant
Class 1 - 10 to 200 psi
Class 2 - 200 psi and above

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

FSC 9330

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SPECIFICATIONS

Federal

L-P-406	Plastics; Organic; General Specifications, Test Methods

Military

MIL-P-9400	Plastic Laminate Materials and Sandwich Construction, Glass Fiber Base, Low Pressure Aircraft Structural, Process Specification Requirements

STANDARD

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Materials.- Materials shall be as specified herein. When materials which are not specifically designated are used, such materials shall be subject to the approval of the procuring activity.

3.2.1 Asbestos felts or mats.- The asbestos felts or mats shall be composed of numerous fibers, unwoven, but randomly distributed. The felts or mats shall conform to table I. These reinforcing sheet materials, after impregnation with resin, shall be uniform and contain no defects or contaminants. Sheet materials approximately 10 mils in thickness are referred to as "felts." Sheet materials with thicknesses greater than 15 mils are referred to as "mats." Composition of felts and mats for the same grade are identical.

3.2.2 Resin.- The resin shall be of the phenolic or modified phenolic thermosetting laminating type. The component materials of the resin shall be such that the cured resin shall not be corrosive to metals. The resin shall be in a form suitable for impregnating or may be already impregnated in asbestos felts or mats (dry lay-up).

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3.3 Process.- A detailed description of the manufacturing and fabricating process and methods of control of manufacturing variables in the form of a titled, numbered, and dated process specification shall be submitted by the contractor to the procuring activity for approval in conformance to Specification MIL-P-9400 and this specification. After approval, the process specification shall form a part of this specification and copies shall be made available by the contractor for use by authorized Government inspectors at the contractor's plant. The materials and processes in production shall conform to the process specification, in addition to the requirements of this specification, and shall not be changed in production without approval of a new superseding process specification. The process specification shall also contain identification of all materials employed and specific requirements for resin content, specific gravity, Barcol hardness, allowable defects (if any), reparable defects (if any), nonreparable defects, and method of repairing any reparable defects. Approval of the fabricator's process specification shall constitute compliance only with minimum acceptable requirements of this specification and Specification MIL-P-9400, and does not obviate any additional requirements as may be specified in any specific procurement of plastic parts. These additional requirements may be specified in the contract, drawings, or detail specifications applicable to the procurement.

3.4 Foreign materials.- No metal staples, paper tape, or other foreign materials shall remain in the finished plastic materials unless allowed by the drawings for the part. Magnetite, an inherent part of chrysotile asbestos, will not be considered as foreign material. In any case, the serviceability of the part shall not be adversely affected.

3.4.1 Laminates.- Finely divided, inert, inorganic fillers may be used in reinforcement of laminates, provided other applicable requirements of this specification are met by the laminates.

3.5 Gaps and laps.- Unless specifically approved by the procuring activity, there shall be no gaps between pieces of asbestos in any lamination. When laps are necessary they shall be laid up with a lap width of at least 1/2 inch but not more than 1 inch, depending on the thickness of the reinforcement used. No two laps shall be superimposed upon each other in laminated material. The total number of laps shall be kept to a minimum.

3.6 Physical properties.- The minimum and maximum values of specific gravity and of resin content, and minimum Barcol hardness value shallbe as specified in the approved contractor's process specification.

3.7 Mechanical properties.- The mechanical properties, in the longitudinal direction, of a 1/8-inch thick parallel laminated flat panel impregnated with the specific resin employed, fabricated in accordance with the approved contractor's process specification, shall conform to tables II and III when tested in accordance with 4.6.2.

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3.7.1 Directional mechanical properties.- The properties as listed in tables II and III are the high properties for laminates using felts or mats. Transverse properties for a parallel lay-up shall produce lowest properties which are at least two-thirds of longitudinal properties. Properties in between longitudinal and transverse are evely proportional between high and low figures. At 45 degrees, properties are at least five-sixths of longitudinal. When lay-up is cross plied at 90 degrees, isotropic properties will be obtained which will be five-sixths of the longitudinal properties listed for parallel lay-ups in tables II and III.

TABLE I

Asbestos base material construction and physical properties

Grade 1/	Form	Weight per 100 sq ft (lbs)	Binder content by weight (percent)	ASTM grade minimum (asbestos) 2/	Thickness (inches) 3/
H	Felt	2.2	3 to 5	AAAA	0.010
HH	Mat	4/	3 to 5	AAAA	0.20

1/ Grade: High.

2/

ASTM grade	Asbestos content percent by weight
AAAA	99 and 100

3/ Thickness is based on relatively no pressure--when fabricating laminate thickness per ply is approximately one-third of the value listed.

4/ Tolerance on weight of felt is ±5 percent. Mats available in different weights up to approximately 30 oz/sq ft.

3.8 Dimensions and make-up of laminates.- The thickness, dimensions, and make-up of the laminate, comprising the number of plies of the various asbestos felts or mats, their position and direction, type of resin and other specific information for the asbestos base, phenolic resin, laminated plastic materials shall be specified in drawings, specifications, or contracts for the parts.

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TABLE II

Longitudinal mechanical properties of types I and II low or high pressure laminated (0.125 ±0.010 inch thick)

Asbestos base, phenolic resin plastic at standard conditions 1/								
Grade	Tensile strength (thousands of psi) Method 1011 2/		Compressive strength (thousands of psi) Method 1021 2/		Flexural strength (thousands of psi) Method 1031 2/		Flexural modulus of elasticity (initial), (Millions of psi) Method 1031 2/	
	Standard cond.	Wet cond.	Standard cond.	Wet cond.	Standard cond.	Wet cond.	Standard cond.	Wet cond.
Class 1								
H or HH	38	36	18	16	40	35	3.0	2.8
Class 2								
H or HH	45	40	20	18	50	45	5.0	4.6

1/ Properties listed are for parallel lay-up panels (see 3.7.1).

2/ Tests are to be performed in accordance with 4.6.2. All values are minimum for the average of 5 specimens. Method numbers refer to Specification L-P-406.

3.9 Overlays.- Asbestos base, low or high pressure, phenolic resin, laminated plastic materials shall not be made with a gel resin overlay, integrally molded or otherwise, or with any other such thick resin surface, or any other surfacing material, unless such overlay or surfacing is specifically approved by the procuring activity.

3.10 Workmanship.-

3.10.1 Except as otherwise specifically approved by the procuring activity, the plastic materials, in those areas of the part which are critical areas structurally, or aerodynamically, shall be uniformly smooth and free from uncured or unbonded areas, gaps, cracks, holes, blisters, resin pockets, areas lacking resin,

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tackiness, excess surface resin, laps, wrinkles, delamination, air or gaspockets, patches, porosity, and other similar defects. The material shall be essentially void free. The critical structural and aerodynamic areas shall be as specified in drawings, specifications, or contracts for the parts.

TABLE III

Longitudinal mechanical properties of type II low or high pressure laminated (0.125 ±0.010 inch thick)

Asbestos base, phenolic base plastic at elevated temperatures 1/										
Property	Flexural strength (thousands of psi)				Flexural modulus of elasticity (initial) (millions of psi)				Compressive strength (thousands of psi)	Tensile strength (thousands of psi)
Test Method No. 2/	1031				1031				1021	1011
Temp. at test and exposure	500° F			700° F	500° F			700° F	500° F	500° F
Time at exposure, hours	1/2	100	192	5	1/2	100	192	5	1/2	1/2
Class 1										
Grade H or HH	35	33	28	15	3.0	2.6	2.2	1.8	13	32
Class 2										
Grade H or HH	45	40	35	25	4.6	4.4	4.0	3.0	15	34

1/ Properties listed are for parallel laminated panels.

2/ Tests are to be performed in accordance with 4.6.2. All values are minimum for the average of 5 specimens. Method numbers refer to Specification L-P-406.

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3.10.2 The allowable defects in the plastic material of the parts shall consist of those included by the contractor as a part of the process specification and then specifically as approved by the procuring activity. These allowable defects shall be fully described in the process specification at to type, size, number, extent, spacing, etc.

3.10.3 The reparable defects, if any, in the plastic material of the parts shall consist of those which can be repaired without affecting the serviceability of the part, and shall be only as specified in the approved process specification. The reparable defects and repair methods shall be fully described in the process specification. Allowable defects shall not affect the serviceability of the part and shall be only as specified in the approved process specification.

3.10.4 Defects greater than those listed in the process specification as allowable or reparable shall be cause for rejection.

3.10.5 Where the requirements as to defects, as stated in the orders, contracts, drawings, or specifications for specific parts are more severe than those in the approved process specifications, the requirements of the orders, contracts, drawings, or specifications for the specific parts shall govern.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of the laminated plastic materials shall be classified as follows:

(a) Preproduction testing (4.3)
(b) Acceptance tests (4.4)

4.3 Preproduction testing.-

4.3.1 Preproduction test samples.- Prior to production of parts on contract, a sample flat sheet 0.125 ±0.010 inch thick shall be low- or high-pressure laminated in accordance with the approved process specification for each of the

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types of resins and felts or mats to be employed in the production of parts. The sample sheets shall be made up in this way and tested separately, in the longitudinal direction, regardless of the continuity of felt or mat and type of lay-up of specific parts. More than one sheet may be made and used for the special tests provided that they are all comparable as to material, process, specific gravity, resin content, and Barcol hardness. Each sample sheet shall be subjected to the preproduction tests specified in 4.3.3.

4.3.2 The preproduction tests need not be repeated for new orders or different parts provided the materials and processes have not been changed and a certified statement to this effect is furnished to the procuring activity with a copy of the report of the tests made. However, in these cases where the procuring activity considers it necessary in order to obtain material conforming to this specification, the right is reserved to require the performance of any of the tests at any time (see 6.2.1).

4.3.3 Preproduction tests.- Preproduction tests shall consist of all tests specified in 4.6.

4.4 Acceptance tests.-

4.4.1 Lot.- A lot of resin, felt, or mat shall be defined as the maximum amount of resin, felt, or mat representative of the same manufacturer's batch or shipment.

4.4.2 Sampling.-

4.4.2.1 Asbestos felt and mat.- A random sample of felt or mat shall be selected from each lot in accordance with Standard MIL-STD-105 at Inspection level III and an AQL of 4.0 percent defective. Each lot of asbestos felt or mat shall be unrolled so that at least 6 yards may be visually inspected for compliance with this specification.

4.4.3 Tests (acceptance).-

4.4.3.1 Resin.- A random sample from each lot of resin shall be inspected for viscosity, gel time, and impurities.

4.4.3.2 Laminates.- From each lot of resin and felt or mat to be used in the fabrication of parts, at least one sample flat sheet 0.125 ±0.010 inch thick shall be laminated in accordance with the approved process specification. Unless otherwise specified herein, five specimens for each test shall be taken from each sample sheet. The acceptance tests shall consist of specific gravity (4.6.1.1), asbestos content (4.6.1.2), Barcol hardness (4.6.1.3), and either ultimate flexural strength dry (4.6.2.3) or tensile strength dry (4.6.2.1) and compression strength edgewise wet (4.6.2.2).

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4.4.3.3 Part.- Each sample molded of the plastic material shall be inspected to determine compliance with the requirements of the approved process specification and Specification MIL-P-9400. This inspection shall be made before any coating or finish of any kind is applied to the molded part.

4.4.4 Rejection.- In case of failure of the sample to pass the specific tests, an additional sample representative of the same lot shall be tested. If this sample fails the specified tests, the material shall be rejected.

4.5 Test conditions.-

4.5.1 Standard test conditions.- Unless otherwise specified, the samples to be tested shall be conditioned for 4 days at 23.4° ±1.1° C (73.4 ±2° F) and 50 ±4 percent relative humidity.

4.5.2 Wet conditions.- Wet conditions shall be 2 hours immersion in boiling distilled water. The specimens shall be cooled in water. The tests shall be made wet immediately after removal of the specimen from the water. In case of any questions about the validity of the test results, wet conditions shall be 30 days soaking in distilled water at room temperature, and testing wet immediately after removal of the specimen from the water. Results determined under these conditions shall be final.

4.5.3 Exposure at 500° F.- Specimens shall be exposed for 1/2 hour at a temperature of 500° ±5° F in the test chamber previously heated to the temperature of test and tested immediately at the same temperature. Specimens for the long term aging tests may be conditioned in a circulating air oven maintained at 500° ±5° F for a period of 100 and 192 hours then transferred to the test chamber (previously heated to the temperature of test) and tested after 1/2 hour at the same temperature.

4.5.4 Exposure at 700° F.- Specimens shall be exposed for 5 hours at a temperature of 700° ±5° F in the test chamber previously heated to the temperature of test and tested immediately at the same temperature.

4.6 Test methods.-

4.6.1 Physical properties.-

4.6.1.1 Specific gravity.- Specific gravity shall be determined in accordance with Method 5011 or 5012 of Specification L-P-406.

4.6.1.2 Asbestos content.- Asbestos content of asbestos laminates shall be determined by the following procedure: At least 3 specimens, 1-inch by 1-inch by the thickness of the sample, shall be cut from the laminate. All loose particles and projecting fibers shall be cleaned from the specimens. Each specimen

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shall be weighed on an analytical balance in a previously weighed crucible. Crucible and specimen shall then be placed in a cold muffle and ignited to constant weight at 900° to 1,000° F. The ignition at 900° to 1,000° F requires from 2 to 6 hours. At the conclusion of this ignition, the specimen shall be heated for at least 1 hour at 1,500° F. The 1,500° F exposure is required to obtain complete loss of water of crystallization in asbestos which amounts to 14 percent. The residue and crucible shall be allowed to cool to room temperature in a desiccator and reweighed on an analytical balance.

Asbestos content, by weight, percent =

$$\frac{W-A}{0.86W} \times 100$$

Where: W = initial weight sample
A = loss of weight

4.6.1.3 Barcol hardness.- Barcol hardness shall be obtained by direct reading on the Barcol impressor. A minimum reading of 55 will be acceptable.

4.6.2 Mechanical properties.-

4.6.2.1 Ultimate tensile strength.- The tensile tests shall be conducted in accordance with Method 1011 of Specification L-P-406 using type II specimens.

4.6.2.2 Compression, edgewise.- Edgewise compressive tests shall be conducted on 0.125 ±0.010 inch thick specimens in accordance with procedures specified in Method 1021 of Specification L-P-406. The tests may be run on thicker material having the same asbestos content and specific gravity as the panel from which the other samples are taken as long as the test conditions are comparable. When a thicker material is used, the test report shall include the thickness of the laminate, specific gravity, and resin content of material tested.

4.6.2.3 Flexural properties, ultimate strength and modulus of elasticity, flatwise.- The flexural test shall be conducted in accordance with Method 1031 of Specification L-P-406.

5. PREPARATION FOR DELIVERY

5.1 The packaging, packing, and marking requirements shall be as specified in the specifications, drawings, or contracts for the parts fabricated from the plastic material.

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NOTES

6.1 Intended use.- The asbestos base, low or high pressure, phenolic laminated plastic material is intended for use in aircraft and missile structural parts, such as wing tips, stabilizers, and fairings, and for use in other applications. All exterior aircraft plastic parts are considered structural parts.

6.1.1 The mechanical properties specified in this specification are for parallel laminated materials. When the materials are used cross-laminated or in other lay-up, or in combinations of lay-ups or materials, the mechanical properties will vary, and for design use may be calculated in accordance with available procedures approved by the procuring activity, or determined empirically. Mechanical properties also may vary with the direction of loading and may also be calculated with other procedures approved by the procuring activity or determined empirically. However, tests are still to be conducted in accordance with this specification.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this or the detail specification, as applicable.
(b) Type and class of material required (see 1.2).
(c) Identification of plies as to felt or mat number required (see 3.2.1).
(d) Process specification details (see 3.3, 3.7, and 3.10).
(e) Type of resin (see 3.6), and overlays (see 3.9).
(f) Make-up of thickness as to number of plies and lay-up, and thickness of material required (see 3.8).

6.2.1 The procuring activity will determine the necessity for a preproduction sample for different parts or when new orders to the same contractor(s) at the same location(s) follow his manufacture of previously satisfactory material covered by this specification. Preproduction tests may be waived, provided the materials and processes have not been changed and the contractor(s) furnishes the procuring activity a certified statement to this effect together with a copy of the results of tests made (see 4.3.2).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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